

Azure AI Search

Cognitive search & retrieval engine for RAG, enterprise search, and knowledge mining

Overview

Azure AI Search (formerly Azure Cognitive Search) is a fully managed search-as-a-service platform that indexes structured and unstructured content, applies AI enrichment via skillsets, and exposes full-text, vector, and hybrid query capabilities. It is the retrieval backbone most commonly paired with Azure OpenAI in Retrieval-Augmented Generation (RAG) architectures.

Core Building Blocks

	Purpose
Data Source	Connection to source data (Blob, ADLS Gen2, SQL, Cosmos DB) for indexer-based ingestion
Index	Schema defining fields, types, attributes (searchable, filterable, facetable, retrievable)
Indexer	Scheduled/on-demand pipeline that pulls data, runs skillset, writes to index
Skillset	Chain of AI enrichment steps (OCR, entity recognition, embeddings, custom skills)
Knowledge Store	Persists enriched data as JSON/tables in Storage for downstream analytics
Semantic Ranker	Re-ranks results using deep learning models for improved relevance

Query Capabilities

- **Full-text search** — Lucene query syntax with scoring profiles and analyzers
- **Vector search** — HNSW-based ANN search over embeddings for semantic similarity
- **Hybrid search** — Combines keyword (BM25) + vector search with Reciprocal Rank Fusion

- **Semantic ranking** — L2 re-ranker improves top-N result ordering using language understanding
- **Filters & facets** — OData filter expressions on filterable fields for precise narrowing

RAG Architecture Pattern

Typical flow: Document ingestion (Blob) → Indexer + Skillset (chunking, embedding via Azure OpenAI text-embedding model) → Vector + hybrid index → Query time: user prompt embedded → hybrid search retrieves top-k chunks → chunks injected into Azure OpenAI chat completion prompt as grounding context.

Security & Access Control

- **Private Endpoints** — isolate the search service from public internet, integrate with VNet
- **Managed Identity** — indexer connects to data sources without stored credentials
- **RBAC roles** — Search Service Contributor, Search Index Data Contributor/Reader (separates admin vs. data-plane access)
- **Customer-managed keys (CMK)** — encrypt index content at rest with Key Vault-held keys
- **Document-level security** — security trimming via filters mapped to Entra ID group claims
- **IP firewall rules** — restrict data-plane and control-plane access by CIDR range

Sizing & Tiers Quick Reference

	Notes
Free	3 indexes, shared infra — dev/testing only
Basic	Dedicated, 15 indexes, good for small production workloads
Standard S1-S3	Scales partitions/replicas independently; S3 HD for many small indexes
Storage Optimized L1/L2	Large-scale, cost-optimized for high-volume knowledge mining

Common Pitfalls

	Fix
Chunking too large for embedding model context	Chunk 300-800 tokens with overlap; test retrieval precision empirically
Mixing vector-only search without keyword fallback	Use hybrid search; pure vector search underperforms on exact-match/rare terms
No security trimming on multi-tenant index	Apply filters mapped to Entra ID group claims at query time, not just UI-level restriction
Re-indexing full corpus on every change	Use indexer change tracking / incremental high-watermark strategy
Ignoring semantic ranker for ambiguous queries	Enable semantic ranking on Standard+ tier for materially better top-N relevance

Interview / Exam Quick Hits

- Replicas scale **query throughput and HA**; partitions scale **storage and indexing throughput**
- Vector search uses **HNSW** algorithm; supports quantization to reduce memory footprint
- Skillsets can call **custom Web API skills** (e.g., Azure Functions) for bespoke enrichment logic
- Semantic ranker is a paid add-on capability, priced per query on Standard tier and above
- Indexers can run **incrementally** using change tracking columns or Blob soft-delete/high-watermark